MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

1/ TYPE 8966

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, ceramic-metal
See figure 1
Mounting position: Vertical, base down or up
Weight: 5.5 pounds (2.5 kg) nominal

ABSOLUTE RATINGS: F = 150 MHz

Parameter: Unit:	Ef Vac	Eb kVdc	Ec3 kVdc	Ec2 Vdc	Ec1 Vdc	Ib Adc	Pg1 W	Pg2 W	Pg3 W	Pp W	Anode core & seal T °C	Cooling --- (Note 1)
Maximum:												
C Teleg:	9.0±5%	7.0	+1.0 -0.5	1,000	-500	2.0	50	175	100	4,000	250	---
C Telep: (anode mod)	9.0±5%	5.0	+1.0 -0.5	800	-500	1.4	50	175	100	2,650	250	---
Class AB:	9.0±5%	7.0	+1.0 -0.5	1,000	---	2.0	50	175	100	4,000	250	---
TEST CONDITIONS:	9.0	2.0	0	750	Adj	1.0	---	---	---	---	---	Note 2

GENERAL:

Qualification - Required

1/ Formerly 8966/5CX3000A

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	39.0	44.0	Aac
1231	Peak emission	-	eb = ec3 = ec2 = ec1 = 2.5 kv	0.65	II	is	40	---	a
1261	Electrode voltage (1) (grid)	-		0.65	II	Ec1	-90	-125	Vdc
1266	Total grid current	-		0.65	II	Ic1	---	-20	µAdc
1256	Electrode current (screen)	-		0.65	II	Ic2	0	+75	mAdc
1266	Primary grid emission (control grid)	-	Pg1 = 60 W or Ic1 = 600 mAdc; t = 15; g2, g3, and p floating	0.65	II	Isg1	---	-50	µAdc
1266	Primary grid emission (screen grid)	-	Ec1 = 0; Pg2 = 210 W or Ic2 = 550 mAdc; t = 15; g3 and p floating	0.65	II	Isg2	---	-150	µAdc
	Quality conformance inspection, part 2								
1372	Current division (method B, short pulse)	-	Eb = Ec2 = 850 Vdc; Ec1 = -300 Vdc; egk/ib = 4.5a	---	---	egk ic2	--- ---	0 1.0	v a
1331	Direct-interelectrode capacitance	-	Ground cathode connection	---	---	Cin Cout Cg1p	125 18.0 ---	145 24.0 0.60	pF pF pF
			Ground grid connection	---	---	Cin	55.0	67.0	pF
1261	Electrode voltage (2) (grid)	-	Eb = 4,000 Vdc; Ec2 = 850 Vdc; Ec1/Ib = 1.0 mAdc	---	---	Eco	---	-300	Vdc
	Quality conformance inspection, part 3								
---	Service-life guarantee	3		---	---	---	---	---	---
2204	Linear amplifier power output and distortion (1-tone)	4	Class AB amp; Eb = 6,000 Vdc; Ec2 = 850 Vdc; Ec1/Ibo = 500 mAdc; F = 2.0 MHz (min); Eg1/Po = 5,500 W (useful); RL = 2,350 ±50 ohms; Rg = 1,000 ohms (max); anode tank Q = 15±2	---	---	3rd IM 5th IM	-32 -37	--- ---	dB dB
1042	Shock, specified pulse	4	No voltages applied; accel = 15G peak (min); shock = 11 ms half-sine; total impacts = 6 (3 each X and Z axes)	---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Shock, specified pulse end points:								
1261	Electrode voltage (1) (grid)	-		---	---	Ec1	-90	-125	Vdc
1266	Total grid current	-		---	---	Ic1	---	-25	µAdc
1032	Vibration, mechanical	4	No voltages applied; accel = 2G peak (min); F = 10 to 50 Hz, ascending only; sweep t = 3 to 8 minutes; 1 sweep each X and Z axes	---	---	---	---	---	---
---	Vibration, mechanical end points:								
1261	Electrode voltage (1) (grid)	-		---	---	Ec1	-90	-125	Vdc
1266	Total grid current	-		---	---	Ic1	---	-25	µAdc

NOTES:

1. Minimum airflow requirements for incoming air at 50°C maximum at sea level, for operation below 30 MHz, are shown. Additional cooling may be required for operation above 30 MHz. In all applications, an air-system socket, such as the EIMAC SK-1420 (or equivalent) and an air chimney, such as the EIMAC SK-1426 (or equivalent) should be used. Air should be directed in a base-to-anode direction, and the values shown are for the tube and the SK-1420/SK-1426 combination with air so directed. In all cases sufficient air must be used to prevent operation with the anode core and/or seal temperatures above the maximum rating, and where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial. Air should be applied before or simultaneously with the application of electrode voltages, including the filament, and may be removed simultaneously with them.

Anode dissipation	Airflow (cfm)	Approximate pressure drop (In. H_2O)
2,000 W	45	0.38
3,000 W	68	0.75
4,000 W	96	1.35

2. In all electrical tests involving application of filament voltage an air-system socket and chimney may be used and forced-air cooling is permissible.

3. The tube manufacturer warrants the tube for 1 year from date of shipment, or 1,000 hours of filament life, whichever first elapses. This warranty applies only when the tube is operated within the maximum ratings (see "Absolute Ratings" of MIL-E-1). A defective tube shall either be replaced, or at the option of the manufacturer, a credit shall be made in the amount of the original purchase price prorated on the basis of 1,000 hours of "filament-on" time.

4. Testing shall be performed every 6 months, with sampling as follows:

$n_1 = 4$ $c_1 = 0$
$n_2 = 4$ $c_2 = 1$; where c_2 represents the total allowable failures for the first and second samples combined.

Separate samples may be used at the option of the manufacturer. None of the listed tests shall be considered destructive except in case of failure. In the event of failure after double sampling, that specific test shall become quality conformance inspection, part 2, AQL 6.5, inspection level S3. After three consecutive successful submissions, the testing may revert to the quality conformance inspection, part 3 test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

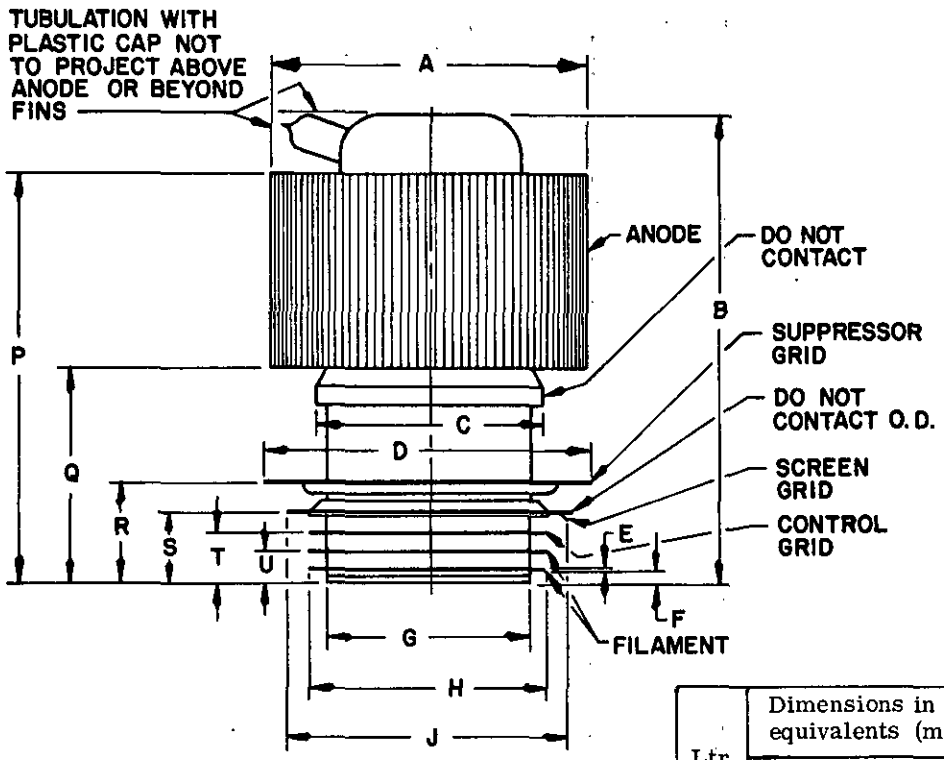
User activities:
Navy - AS, OS, MC, CG

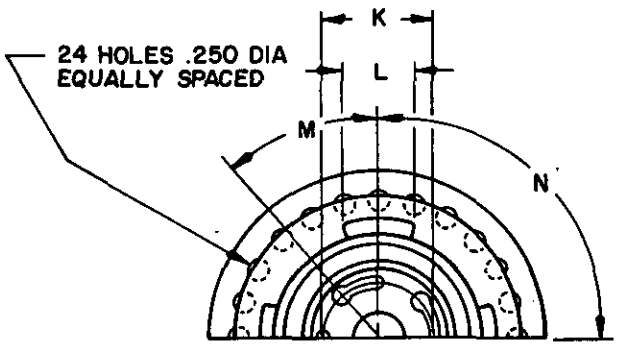
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N098)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.





Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	4.585 (116.46)	4.625 (117.47)
B	6.638 (168.60)	6.838 (173.68)
D	4.615 (117.22)	4.635 (117.73)
E	.018 (.46)	.025 (.63)
H	3.490 (88.65)	3.525 (89.53)
K	1.557 (39.55)	1.567 (39.80)
L	1.050 (26.67)	1.100 (27.94)
M	39°	41°
N	89°	91°
P	5.900 (149.86)	6.100 (154.94)
R	1.385 (35.18)	1.485 (37.72)
S	.930 (23.62)	1.010 (25.65)
T	.700 (17.78)	.730 (18.54)
U	.430 (10.92)	.460 (11.68)
Quality conformance inspection, part 3 (See note)		
C	3.380 (85.85)	3.450 (87.63)
F	.160 (4.06)	.180 (4.57)
G	2.985 (75.82)	3.025 (76.83)
J	4.000 (101.60)	4.175 (106.04)
Q	3.050 (77.47)	3.150 (80.01)

NOTE: Dimensions shall be checked every 6 months, with sampling as follows:

n1 = 4 c1 = 0 where c2 represents the total allowable failures for the first and second samples combined.

n2 = 4 c2 = 1;

Separate samples may be used at the option of the manufacturer. None of the listed tests shall be considered destructive except in case of failure. In the event of failure after double sampling; that specific test shall become quality conformance inspection, part 2; after three consecutive successful submissions, the testing may revert to the quality conformance inspection, part 3 tests.

FIGURE 1. Outline drawing of electron tube type 8966.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:32Z
Check the source to verify that this is the current version before use.